TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*PASTE2POWER – Infrastructure-Independent, Low-Noise Backup Power Using POWERPASTE-to-Hydrogen and PEM Fuel Cells (≤10 kW; modular roadmap to 50 kW)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*New & Renewable Energy*	*Fuel cell*
Concept of Proposed Technology	*We propose a stationary (and portable) backup power generator in which POWERPASTE and water are stored separately and converted on demand into hydrogen (H_2) in a compact generator. The hydrogen can be supplied directly to a PEM fuel cell that produces clean electricity with very low noise and zero local combustion emissions. The system targets infrastructure-poor, environmentally sensitive, or reliability-critical sites where diesel is impossible, unwanted, or economically unattractive over the full lifecycle.* *The system has zero local combustion emissions and very low noise; lifecycle emissions depend on POWERPASTE production and logistics, and will be quantified in WP6.* *Backup generators typically run rarely, yet demand high reliability and low standby burden. POWERPASTE's value is not only "fuel cost per kWh", but total cost of ownership (TCO) driven by:*			

	• *Long shelf life (minimum ≥2 years) and no self-discharge, enabling long-term readiness without fuel turnover.* • *Simplified logistics: transport and storage without high-pressure cylinders or cryogenic supply chains, reducing operational friction and specialised training needs.* • *On-demand hydrogen generation matched to load, enabling efficient part-load operation and fast refuelling via cartridge/container exchange or bulk refill concepts.* *Even at today's early-market pricing, POWERPASTE-based hydrogen can be approximately €20 – €30 (₩34,000 – ₩52,000) per kg H_2. At ≥45% DC system efficiency (hydrogen generator integrated with a fuel cell), this corresponds to roughly €1.3 – €2.0 (₩2,200 - ₩3,400) per kWh-electric (excluding CAPEX). This proposal targets cases where diesel's apparent cheapness collapses once maintenance, periodic inspection, fuel ageing/contamination, logistics, noise limits, air-quality restrictions, and reliability costs are included. WP6 will quantify TCO versus diesel and batteries for selected Korean duty cycles.* *Primary markets are SME-operated or SME-served sites needing quiet, clean, dependable backup in the 0.5–10 kW class, with a clear path to 10–50 kW via modularisation:* • *Telecom shelters / remote repeater sites, traffic and rail signalling, disaster-response depots* • *Environmental monitoring / protected areas where combustion is restricted* • *Industrial standby power for critical control systems where maintenance visits are costly* • *Remote islands, mountainous infrastructure, temporary worksites requiring ultra-low noise* *Fraunhofer IFAM Dresden is pursuing reactor and hydrogen-generator scaling in the 5–50 kW class (SmartPASTE project), aligning with this proposal's upper power boundary and reducing scale-up risk.*
Rationale for Joint R&D Project	*This technology sits at the intersection of materials know-how (POWERPASTE + hydrolysis reactor control) and Korean-strength industrialisation (power electronics, fuel-cell integration, manufacturing, certification, and service networks). A joint program is an efficient route to:* • *Technology localisation (Korean codes, safety practices, supply chain fit)* • *Korea-based pilot deployments with credible reference customers* • *Rapid product development beyond a laboratory demonstrator* *Fraunhofer IFAM Dresden is the inventor and developer of POWERPASTE and we position it explicitly for fuel cell power generators, highlighting long shelf life, logistics advantages, and the targeted 1 kW to 10 kW application range.* *Fraunhofer IFAM has an established history of Korea collaboration, including a research agreement (MoU) with Hanyang University to promote education, research, and industry cooperation. IFAM Dresden is part of a broader strategic cooperation with Korea's energy research ecosystem:*

	Fraunhofer IFAM Dresden and KIER signed a cooperation agreement to intensify exchange on hydrogen and renewables. *TIPA's mission is explicitly to increase SME competitiveness through technology innovation and coordinated R&D support, making an SME-led product development project a natural fit. Additionally, Fraunhofer institutes in Dresden have demonstrated practical Korea-funding interfaces, which reduces administrative friction and accelerates partnership execution.*
Key Words	1. *POWERPASTE* 2. *Fuel cell backup power* 3. *On-demand hydrogen generation* 4. *Low-TCO* 5. *Off-grid energy*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	TRL 4	End of the Project	TRL 6-7

R&D Objectives

Possible Primary objective (MVP):

Design, build, and validate a Korea-deployable 5 kW-class POWERPASTE-to-PEMFC backup generator prototype with:

- *Load-following hydrogen generation*
- *Outdoor-capable storage concept (POWERPASTE + water managed separately)*
- *Refuelling concept suitable for non-specialist operators*
- *Safety architecture aligned to Korean certification expectations*
- *Field demonstration under a representative duty cycle (standby + start + dynamic load)*

Possible Secondary objective (Scaling, modularisation):

- *Define a modular architecture enabling a product family: 1 kW / 5 kW / 10 kW*
- *Deliver an engineering path to ≤50 kW, e.g. via parallel modules, consistent with IFAM Dresden's 5–50 kW scale-up activities.*

Depending on the final SME need the objectives of the project can change.

R&D Target Specifications

Category	***Possible target specification***
Electrical output	*5 kW continuous, scalable module concept to 10 kW, roadmap to ≤50 kW*
Electrical efficiency	*≥ 45% DC (fuel cell system level, nominal load)*
Start / response	*Start to rated power ≤60 s; step response ≤5 s to 50% load change (buffer battery allowed)*
Hydrogen delivery	*On-demand, load-following H_2 generation; stable supply designed for PEM fuel cell requirements*
Operating pressure	*Low-pressure hydrogen generation and feed (target <5 bar system domain)*
Noise	*"Low-noise" design target <45 dBA @ 7 m (site dependent, enclosure option)*
Emissions at point of use	*Zero local combustion emissions during operation (electricity from fuel cell)*
Storage readiness	*POWERPASTE storage with minimum ≥2-year shelf life; water stored separately; outdoor enclosure option*
Environment	*Outdoor operation concept (cold-weather and heat management); freeze-protection concept for water loop*
Refuelling / replenishment	*Cartridge or container exchange / refill concept; tool-light service; designed for SME operators*
Safety	*Multi-layer safety: dosing control, temperature management, pressure relief strategy, leak detection, safe shutdown, by-product containment*
Maintenance concept	*Backup-optimised: minimal periodic servicing; remote health monitoring option*

Exemplary use case (telecom shelter backup)

Power:	*5 kW*
Runtime target:	*8 h*
Electricity delivered:	*40 kWh*
H_2 required (45% system efficiency):	*~2.7 kg H_2 (1 kg H_2 ≈ 33.3 kWh LHV; electric ≈ 33.3×0.45 ≈ 15 kWh/kg → 40/15 ≈ 2.7 kg)*
POWERPASTE required (indicative):	*~28 kg / 23 Liters (based on measured H_2 content)*
Water required (indicative):	*~50 Liters (based on experimental data)*

	By-product handling method: sealed container capture, take-back and regeneration/recycling pathway defined in WP6
R&D Plan	***Overall duration**: 28–36 months.* ***Project governance**: joint steering committee; monthly technical review; quarterly milestone evaluation.* *Phase 1 (moths 1-6)* ***Work Package 1** – Requirements, use-case definition, and Korean compliance mapping (Months 0–6)* • *Select 1–2 priority Korean use cases (e.g., telecom backup shelter, protected-area monitoring station, industrial control standby)* • *Translate duty cycles into engineering requirements: start behaviour, dynamic load, runtime, maintenance interval* • *Compliance mapping (safety, transport/storage classification, site permitting, electrical interconnect)* • *Basic engineering of the system* • *Deliverables: User requirement spec (URS), compliance checklist, pilot site definition.* *Phase 2 (moths 7-36)* ***Work Package 2** – POWERPASTE hydrogen generator engineering for 5 kW continuous (Months 7–15)* • *Reactor/hardware design for controlled hydrolysis, stable H_2 output* • *Thermal management and safe by-product handling concept* • *Dosing and control strategy for dynamic operation* • *Design for manufacturability (DfM) with Korean SME feedback* ***Milestone M1** (Month 50): validated sub-systems (dosing, thermal, safety logic) at relevant throughput.* ***Work Package 3** – Fuel cell system integration and power electronics (Months 7–19)* • *Select PEM fuel cell stack and balance-of-plant; define operating window matched to H_2 generator behaviour* • *DC output conditioning; optional inverter / UPS-like interface* • *Buffer battery strategy for transients (optimise fuel cell sizing and lifetime)* ***Milestone M2** (Month 19): integrated lab prototype producing stable power under dynamic load profile.* ***Work Package 4** – Storage, packaging, and field-service design (Months 10–22)* • *Outdoor enclosure and environmental protection* • *Replenishment workflow (non-specialist operation)*

	• *Diagnostics and remote monitoring interface (optional, strong Korean market pull)* ***Milestone M3*** *(Month 22): "field-ready" MVP package.* ***Work Package 5*** *– Verification testing, safety validation, and reliability growth (Months 15–28)* • *Functional safety tests (fault injection: dosing fault, sensor fault, water feed interruption)* • *Reliability growth testing under start/stop cycling, standby periods, temperature variation* • *Pre-certification testing plan aligned with Korean standards pathway* • *Safety concept & fault tree* • *Preliminary hazard analysis (PHA) / HAZOP-lite* ***Milestone M4*** *(Month 28): verified MVP meeting spec thresholds and safety cases.* ***Work Package 6*** *– Korea field demonstration and commercial readiness package (Months 20–36)* • *Pilot deployment in Korea with logged performance* • *Cost model refinement (CAPEX + OPEX + logistics), service plan, supply chain plan* • *Roadmap to 10 kW and modular scale-up (≤50 kW) backed by measured data* ***Final deliverables****: demonstrated MVP + data package + certification pathway + product development plan.* ***Risk management*** *(embedded in the work plan):* • *Thermal excursions / reaction control: mitigated by dosing strategy, heat removal design, and safety interlocks* • *Hydrogen purity stability: mitigated by system conditioning and monitoring concept compatible with fuel cell operation* • *Cold weather operation: mitigated by enclosure + water freeze-protection strategy* • *Regulatory uncertainty: mitigated early via WP1 compliance mapping and test plan alignment*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *POWERPASTE know-how transfer for safe system integration (materials behaviour, handling windows, dosing philosophy)* • *Design and optimisation of the hydrogen generator for dynamic load following and 5 kW-class throughput* • *Safety architecture and validation methodology; by-product management concept* • *Scale-up roadmap alignment with ongoing 5–50 kW class reactor development activities* • *Joint testing, data interpretation, and preparation of the technical documentation package for certification readiness* • *Provision of POWERPASTE for field tests*
Expected R&R of Korean SME Partner	• *System engineering leadership for product-grade integration (PEMFC selection, BOP, power electronics, enclosure)* • *Korean compliance execution: documentation, test coordination, certification interface* • *Manufacturing engineering, supply chain localisation, cost-down design iterations* • *Pilot site access and field operation support in Korea; service model definition* • *Commercialisation plan: target customers, channel strategy, after-sales readiness*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*1) Exemplar technology outcome:* • *5 kW-class MVP demonstrated with real duty cycles and a complete engineering data package* • *Modular blueprint for a ≤10 kW product line and a credible pathway to ≤50 kW via parallel modules* *2) SME competitiveness and export potential:* • *A differentiated Korean SME product positioned where batteries and diesel struggle: long standby readiness, quick replenishment, low noise, clean operation, and simplified logistics* • *Export-ready reference case: "Korea-validated clean backup power" for global infrastructure markets* *3) National and societal benefit:* • *Increased resilience of critical infrastructure (disaster readiness, remote sites, essential services)* • *Practical contribution to clean energy adoption without requiring build-out of high-pressure or cryogenic hydrogen infrastructure* *4) Strategic partnership impact:* • *This project builds on proven Korea–Fraunhofer collaboration patterns (e.g., IFAM Dresden's MoU with Hanyang University and Fraunhofer–KIER cooperation including IFAM Dresden Branch), strengthening a long-term pipeline from joint R&D to industrial deployment.*

6. References

(1) Tegel, M.; Schöne, M.; Kieback, B.; Röntzsch, L. "An efficient hydrolysis of MgH_2-based materials." International Journal of Hydrogen Energy, 42(1), 183–191 (2017). DOI: 10.1016/j.ijhydene.2016.09.084

(2) Fraunhofer IFAM. "POWERPASTE FOR OFF-GRID POWER SUPPLY" (White Paper, May 2022). https://www.ifam.fraunhofer.de/content/dam/ifam/en/documents/dd/Infobl%C3%A4tter/White_paper_POWERPASTE_final.pdf

(3) Fraunhofer IFAM (Dresden). "POWERPASTE" technology page (key facts, performance and application context).

https://www.ifam.fraunhofer.de/en/Aboutus/Locations/Dresden/HydrogenTechnology/powerpaste.html

(4) Fraunhofer ZESS (Fraunhofer IKTS). "POWERPASTE – chemical hydrogen storage: Hydrolysis" (reaction and key figures).

https://www.zess.fraunhofer.de/en/competencies/hydrogen_technologies/powerpaste-hydrolysis.html

(5) European Patent EP3154900B1. "Composite material for hydrolytically generating hydrogen … and possible applications." https://patents.google.com/patent/EP3154900B1/en

(6) US Patent US10239753B2 (PDF). "Composite material for hydrolytically generating hydrogen … and possible applications." https://patentimages.storage.googleapis.com/87/3f/fa/54a7bc14ab736d/US10239753.pdf

(7) Fraunhofer (Mediathek). "Video: POWERPASTE – Wasserstoffantriebe für Kleinfahrzeuge" (official Fraunhofer video page). https://www.fraunhofer.de/de/mediathek/filme/filme-2021/powerpaste.html

(8) H2FCHannover (YouTube). "Recent Advances in POWERPASTE: Changing the Future of Energy Storage" (Fraunhofer IFAM talk). https://www.youtube.com/watch?v=gTWwrmeulPg

(9) Deutscher Bundestag – Wissenschaftliche Dienste. “Kurzinformation: Wasserstoffpaste als Energiespeicher” (WD-8-021-21). https://www.bundestag.de/resource/blob/831882/d2197fad623a700b3b43e484870db6e3/WD-8-021-21-pdf-data.pdf

(10) HTW Dresden (News). “Pionierarbeit für die Wasserstoff-Technologie” (SmartPASTE context and scale-up direction). https://www.htw-dresden.de/news/pionierarbeit-fuer-die-wasserstoff-technologie

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
X	**Prototype Stage (5-6)**	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization